

ECON1050 Tutorial 10 & 11

Matrix Algebra I & II

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Today's Plan

- ▶ This is our second last tutorial. No tutorial in week 13!
- ▶ Recap: key ideas from Lecture 10 & 11
- ▶ Tutorial questions:
 - ▶ Tute 10: Q2, and Q4
 - ▶ Tute 11: Q1, Q2, and Q3

Recap: Matrices and notation

Definition

A **matrix** is a rectangular array of numbers.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Dimension

A matrix with m rows and n columns has dimension $m \times n$.

Elements

The element a_{ij} is in row i and column j .

Recap: Scalars, vectors, and special matrices

Scalars and vectors

- ▶ A **scalar** is a single number.
- ▶ A **vector** is a matrix with only one row or only one column.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{p}' = [p_1 \quad p_2 \quad \cdots \quad p_n].$$

Special matrices

- ▶ A **square matrix** has dimension $n \times n$.
- ▶ A **diagonal matrix** has all off-diagonal entries equal to zero.
- ▶ An **identity matrix** is a diagonal matrix with ones on the main diagonal.
- ▶ A **zero matrix** has all entries equal to zero.

Recap: Equality, addition, and scalar multiplication

Equal matrices

Two matrices are equal only if they have the same dimension and all corresponding entries are identical.

Addition and subtraction

If $\mathbf{A}$ and $\mathbf{B}$ have the same dimension, then

$$\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}], \quad \mathbf{A} - \mathbf{B} = \mathbf{A} + (-1)\mathbf{B}.$$

Scalar multiplication

For a scalar k ,

$$k\mathbf{A} = [ka_{ij}].$$

Recap: Properties of matrix addition

Addition properties

For conformable matrices:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A} \quad (\text{commutative})$$

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C} \quad (\text{associative})$$

$$\mathbf{A} + \mathbf{0} = \mathbf{A}.$$

Scalar multiplication properties

$$k(\mathbf{A} + \mathbf{B}) = k\mathbf{A} + k\mathbf{B}, \quad (k_1 + k_2)\mathbf{A} = k_1\mathbf{A} + k_2\mathbf{A}.$$

Recap: Transpose of a matrix

Definition

The **transpose** of $\mathbf{A}$ is denoted by $\mathbf{A}'$ or $\mathbf{A}^T$. The columns of $\mathbf{A}$ become the rows of $\mathbf{A}'$.

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \mathbf{A}' = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}.$$

Key properties

$$(\mathbf{A}')' = \mathbf{A}, \quad (c\mathbf{A})' = c\mathbf{A}', \quad (\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'.$$

Recap: Matrix multiplication

Conformability

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and has dimension $m \times p$.

$$\mathbf{A}_{m \times n} \mathbf{B}_{n \times p} = \mathbf{C}_{m \times p}.$$

Element formula

The (i, j) entry of $\mathbf{C} = \mathbf{AB}$ is

$$c_{ij} = \sum_{r=1}^n a_{ir} b_{rj}.$$

This is the inner product of row i of $\mathbf{A}$ and column j of $\mathbf{B}$.

Recap: Matrix multiplication rules

Important warning

In general,

$$\mathbf{AB} \neq \mathbf{BA}.$$

Matrix multiplication is not commutative.

Useful properties

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C},$$

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC},$$

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}.$$

$$(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'.$$

Recap: Powers and special square matrices

Powers

If $\mathbf{A}$ is square, then

$$\mathbf{A}^2 = \mathbf{A}\mathbf{A}, \quad \mathbf{A}^3 = \mathbf{A}\mathbf{A}\mathbf{A}.$$

Idempotent matrix

A square matrix $\mathbf{A}$ is **idempotent** if

$$\mathbf{A}^2 = \mathbf{A}.$$

Symmetric and orthogonal matrices

- ▶ $\mathbf{A}$ is **symmetric** if $\mathbf{A} = \mathbf{A}'$.
- ▶ $\mathbf{A}$ is **orthogonal** if $\mathbf{A}'\mathbf{A} = \mathbf{I}$.

Recap: Vectors, inner product, and orthogonality

Inner product

For two n -vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^n a_i b_i.$$

Norm

The length or norm of $\mathbf{a}$ is

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}.$$

Orthogonality

Two vectors are orthogonal if

$$\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0.$$

Recap: Systems of linear equations

Matrix form

A system of linear equations can be written as

$$\mathbf{Ax} = \mathbf{b},$$

where

- ▶ $\mathbf{A}$ is the **coefficient matrix**,
- ▶ $\mathbf{x}$ is the **vector of variables**,
- ▶ $\mathbf{b}$ is the **vector of constants**.

Example structure

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}.$$

Recap: Elementary row operations

Valid row operations

The following operations do not change the solution set of a system of linear equations:

- ▶ Interchange any pair of rows.
- ▶ Multiply any row by a nonzero scalar.
- ▶ Add any multiple of one row to a different row.

Augmented matrix

For a system $\mathbf{Ax} = \mathbf{b}$, use

$$[\mathbf{A} \mid \mathbf{b}].$$

Gaussian elimination applies row operations to this augmented matrix.

Recap: Gaussian elimination

Goal

Convert the coefficient side of the augmented matrix into a simpler form.

Row-echelon form

$$\begin{bmatrix} 1 & a & b & d \\ 0 & 1 & c & e \\ 0 & 0 & 1 & f \end{bmatrix}.$$

Then use backward substitution.

Reduced row-echelon form

$$\begin{bmatrix} 1 & 0 & 0 & g \\ 0 & 1 & 0 & h \\ 0 & 0 & 1 & i \end{bmatrix}.$$

Then the solution is read directly.

Recap: Inverse of a matrix

Definition

The inverse of a square matrix $\mathbf{A}$ is denoted by $\mathbf{A}^{-1}$ and satisfies

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \mathbf{A}^{-1}\mathbf{A}.$$

Solving a system using the inverse

If

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

and $\mathbf{A}^{-1}$ exists, then

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Recap: Properties of inverse matrices

Key facts

- ▶ Only square matrices can have inverses.
- ▶ Not every square matrix has an inverse.
- ▶ If $\mathbf{A}^{-1}$ exists, it is unique.
- ▶ If $\mathbf{A}^{-1}$ exists, then $\mathbf{A}$ is **non-singular**.
- ▶ A non-singular matrix has linearly independent rows and columns.

Singular matrix

If no inverse exists, $\mathbf{A}$ is called **singular**.

Recap: Inverse by Gaussian elimination

Method

For an invertible $n \times n$ matrix $\mathbf{A}$:

$$[\mathbf{A} \mid \mathbf{I}] \longrightarrow [\mathbf{I} \mid \mathbf{A}^{-1}].$$

Interpretation

The same row operations that transform $\mathbf{A}$ into $\mathbf{I}$ transform $\mathbf{I}$ into $\mathbf{A}^{-1}$.

Recap: Determinants

Definition

A determinant is a scalar number associated with a square matrix.

$$|\mathbf{A}| = \det(\mathbf{A}).$$

Second-order determinant

For

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$|\mathbf{A}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Recap: Determinants and non-singularity

Non-singularity

A square matrix $\mathbf{A}$ is non-singular if and only if

$$|\mathbf{A}| \neq 0.$$

Singularity

If

$$|\mathbf{A}| = 0,$$

then $\mathbf{A}$ has no inverse.

Systems of equations

A system $\mathbf{Ax} = \mathbf{b}$ has a unique solution when $\mathbf{A}$ is non-singular.

Recap: Third-order determinants

Expansion by cofactors

For a 3×3 matrix, the determinant can be found by expanding along a row or column.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow |\mathbf{A}| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

$C_{ij} = (-1)^{i+j}|M_{ij}|$ is Cofactor, where M_{ij} is the minor obtained by deleting row i and column j .

Sarrus' Rule

$$\begin{aligned} |\mathbf{A}| = & + a_{11} \times a_{22} \times a_{33} + a_{12} \times a_{23} \times a_{31} + a_{13} \times a_{21} \times a_{32} \\ & - a_{31} \times a_{22} \times a_{13} - a_{32} \times a_{23} \times a_{11} - a_{33} \times a_{21} \times a_{12} \end{aligned}$$

Recap: Laplace expansion

General formula

For an $n \times n$ matrix $\mathbf{A} = \{a_{ij}\}$, the determinant may be expanded along any row or column:

$$|\mathbf{A}| = \sum_{j=1}^n a_{ij} C_{ij} \quad \text{for any row } i,$$

or

$$|\mathbf{A}| = \sum_{i=1}^n a_{ij} C_{ij} \quad \text{for any column } j.$$

Practical tip

Choose a row or column with zeros to reduce the amount of calculation.

Recap: Properties of determinants

Key properties

► If a row or column contains only zeros, then $|\mathbf{A}| = 0$.

► $|\mathbf{A}| = |\mathbf{A}'|$.

► If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same order, then

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}|.$$

► Interchanging two rows or two columns changes the sign of the determinant.

Recap: More determinant properties

Scaling and row operations

► Multiplying one row or column by k multiplies the determinant by k .

► If $\mathbf{A}$ is $n \times n$, then

$$|k\mathbf{A}| = k^n |\mathbf{A}|.$$

► Adding a multiple of one row or column to another row or column does not change the determinant.

► If one row or column is a multiple of another, then $|\mathbf{A}| = 0$.

Tutorial 10: Q1

Question 1

Evaluate the following integrals using integration by substitution:

$$(1) \int \frac{3x}{(3x^2 - 4)^2} dx, \quad (2) \int \frac{1}{\sqrt{x}} e^{\sqrt{x}} dx.$$

Tutorial Q1: Integration by substitution

Knowledge used

- ▶ Revision from integration: use substitution as the reverse of the chain rule.
- ▶ If $u = g(x)$ and $du = g'(x) dx$, then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

Solution

For

$$\int \frac{3x}{(3x^2 - 4)^2} dx,$$

let $u = 3x^2 - 4$, so $du = 6x dx$. Hence

$$\int \frac{3x}{(3x^2 - 4)^2} dx = \frac{1}{2} \int u^{-2} du = -\frac{1}{2(3x^2 - 4)} + C.$$

Tutorial Q1: Integration by substitution

Knowledge used

- ▶ Revision from integration: use substitution as the reverse of the chain rule.
- ▶ If $u = g(x)$ and $du = g'(x) dx$, then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

Solution

For

$$\int \frac{1}{\sqrt{x}} e^{\sqrt{x}} dx,$$

let $u = \sqrt{x}$, so $du = \frac{1}{2\sqrt{x}} dx$. Hence

$$\int \frac{1}{\sqrt{x}} e^{\sqrt{x}} dx = 2 \int e^u du = 2e^{\sqrt{x}} + C.$$

Tutorial 10: Q2

Question 2

Use the following matrices:

$$\mathbf{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix},$$

$$\mathbf{D} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 4 & 1 & -4 \\ 2 & 1 & 6 \\ -1 & 0 & 3 \end{bmatrix}.$$

Calculate the following matrices where possible:

(a) $\mathbf{A} + \mathbf{B}$; (b) $\mathbf{A} - 5\mathbf{B}$; (c) $\mathbf{B} + 4\mathbf{C}$; (d) $\mathbf{C}'$; (e) $\mathbf{D} + \mathbf{E}$; (f) $(2\mathbf{F})'$.

Tutorial Q2: Matrix addition and scalar multiplication I

Knowledge used

- ▶ Matrices can be added only if they have the same dimension.
- ▶ Scalar multiplication multiplies every element of a matrix.

Solution

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}.$$

$$\mathbf{A} - 5\mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 5 \begin{bmatrix} -4 \\ 0 \end{bmatrix} = \begin{bmatrix} 21 \\ 2 \end{bmatrix}.$$

Tutorial Q2: Matrix addition and transpose II

Knowledge used

- ▶ Check conformability before adding matrices.
- ▶ The transpose swaps rows and columns.

Solution

$$\mathbf{B} + 4\mathbf{C}$$

has **no solution** because $\mathbf{B}$ is 2×1 and $\mathbf{C}$ is 2×2 .

$$\mathbf{C}' = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}' = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}.$$

Tutorial Q2: Transpose operations III

Knowledge used

- Use $(\mathbf{A}')' = \mathbf{A}$.
- Use $(c\mathbf{A})' = c\mathbf{A}'$.

Solution

$$\mathbf{D} + \mathbf{E}' = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 2 \\ -1 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 2 & -1 & 6 \end{bmatrix}.$$

$$(2\mathbf{F}')' = 2\mathbf{F} = \begin{bmatrix} 8 & 2 & -8 \\ 4 & 2 & 12 \\ -2 & 0 & 6 \end{bmatrix}.$$

Tutorial 10: Q3

Question 3

Solve for t :

$$\begin{bmatrix} 2t & 1 \\ -3 & t+1 \end{bmatrix} = 2 \begin{bmatrix} t & 1/2 \\ t+1/2 & -1/2 \end{bmatrix}.$$

Tutorial Q3: Solving from matrix equality

Knowledge used

- ▶ Equal matrices must have identical corresponding elements.
- ▶ Use scalar multiplication before comparing entries.

Solution

$$2 \begin{bmatrix} t & 1/2 \\ t+1/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 2t & 1 \\ 2t+1 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2t & 1 \\ -3 & t+1 \end{bmatrix} = \begin{bmatrix} 2t & 1 \\ 2t+1 & -1 \end{bmatrix}.$$

Compare corresponding elements:

$$-3 = 2t + 1 \Rightarrow t = -2.$$

Also,

$$t + 1 = -1 \Rightarrow t = -2.$$

Therefore, $t = -2$.

Tutorial 10: Q4

Question 4

Use the following matrices:

$$\mathbf{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix},$$

$$\mathbf{D} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 4 & 1 & -4 \\ 2 & 1 & 6 \\ -1 & 0 & 3 \end{bmatrix}.$$

Compute:

$$\mathbf{A}'\mathbf{B}, \quad \mathbf{AB}', \quad \mathbf{AB}, \quad \mathbf{AB}'\mathbf{C}, \quad \mathbf{ED}, \quad (\mathbf{ED})', \quad \mathbf{CDE}.$$

Tutorial Q4: Matrix multiplication I

Knowledge used

- ▶ Matrix multiplication requires inner dimensions to match.
- ▶ The product of a 1×2 matrix and a 2×1 matrix is 1×1 .
- ▶ The product of a 2×1 matrix and a 1×2 matrix is 2×2 .

Solution

$$\mathbf{A}'\mathbf{B} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -4 \\ 0 \end{bmatrix} = -4.$$

$$\mathbf{AB}' = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} -4 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ -8 & 0 \end{bmatrix}.$$

$\mathbf{AB}$ is not conformable.

Tutorial Q4: Matrix multiplication II

Knowledge used

- ▶ Matrix multiplication is associative:

$$(\mathbf{A}\mathbf{B}')\mathbf{C} = \mathbf{A}(\mathbf{B}'\mathbf{C})$$

whenever the products are defined.

Solution

$$\begin{aligned}\mathbf{A}\mathbf{B}'\mathbf{C} &= \begin{bmatrix} -4 & 0 \\ -8 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -4 & -8 \\ -8 & -16 \end{bmatrix}.\end{aligned}$$

Tutorial Q4: Matrix multiplication III

Knowledge used

- ▶ The product of a 3×2 matrix and a 2×3 matrix is 3×3 .
- ▶ Use row-by-column multiplication for each entry.

Solution

$$\begin{aligned} \mathbf{ED} &= \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \\ 3 & 0 & 6 \\ 17 & -5 & 9 \end{bmatrix}. \\ (\mathbf{ED})' &= \begin{bmatrix} -2 & 3 & 17 \\ 1 & 0 & -5 \\ 1 & 6 & 9 \end{bmatrix}. \end{aligned}$$

Tutorial Q4: Matrix multiplication IV

Knowledge used

- ▶ Check dimensions at every step.
- ▶ For **CDE**:

$$(2 \times 2)(2 \times 3)(3 \times 2) \Rightarrow 2 \times 2.$$

Solution

First,

$$DE = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 3 & 0 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 2 & 2 \end{bmatrix}.$$

Then

$$CDE = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 5 & 9 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 13 \\ 3 & -1 \end{bmatrix}.$$

Key takeaways

- ▶ Always check matrix dimensions before adding or multiplying.
- ▶ Matrix addition is componentwise, but matrix multiplication is row-by-column.
- ▶ In general, $\mathbf{AB} \neq \mathbf{BA}$.
- ▶ Transpose reverses product order: $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$.
- ▶ Vectors are special matrices, and their inner product is useful in economics.

Tutorial 11: Q1

Question 1

Consider the system

$$2x + y + z = 0, \quad 4x + 3y + 2z = 2, \quad 2x - y - 3z = 0.$$

1. Write the system in matrix form.
2. Solve the system using Gaussian elimination.

Tutorial Q1: Matrix form

Knowledge used

- ▶ A linear system can be written as $\mathbf{Ax} = \mathbf{b}$.
- ▶ The coefficient matrix contains the coefficients of the variables.

Solution

$$\mathbf{Ax} = \mathbf{b}$$

where

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Tutorial Q1: Gaussian elimination I

Knowledge used

- Use the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$.
- Apply elementary row operations to simplify the system.

Solution

$$[\mathbf{A} \mid \mathbf{b}] = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 4 & 3 & 2 & 2 \\ 2 & -1 & -3 & 0 \end{bmatrix}.$$

First divide row 1 by 2:

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 4 & 3 & 2 & 2 \\ 2 & -1 & -3 & 0 \end{bmatrix}.$$

Then use row 1 to eliminate the first entries in rows 2 and 3: $R_2 \leftarrow R_2 - 4R_1$, $R_3 \leftarrow R_3 - 2R_1$.

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 2 \\ 0 & -2 & -4 & 0 \end{bmatrix}.$$

Tutorial Q1: Gaussian elimination II

Knowledge used

- ▶ Continue row reduction until the system is triangular or reduced.
- ▶ Then use backward substitution.

Solution

Use row 2 to eliminate the second entry in row 3: $R_3 \leftarrow R_3 + 2R_2$.

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -4 & 4 \end{bmatrix}.$$

Divide row 3 by -4 :

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

Thus $z = -1$, and $y = 2$. Substitute into row 1: $x + \frac{1}{2}(2) + \frac{1}{2}(-1) = 0 \Rightarrow x = -\frac{1}{2}$. Therefore,

$$(x, y, z) = \left(-\frac{1}{2}, 2, -1\right).$$

Tutorial 11: Q2

Question 2

Calculate the inverse of the following matrices, if possible, using Gaussian elimination. Check your result by calculating $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

$$(a) \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \quad (b) \begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix},$$

$$(c) \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}.$$

Tutorial Q2(a): Inverse by Gaussian elimination

Knowledge used

- ▶ To find $\mathbf{A}^{-1}$, row-reduce $[\mathbf{A} \mid \mathbf{I}]$ into $[\mathbf{I} \mid \mathbf{A}^{-1}]$.
- ▶ Verify by checking $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

Solution

For

$$\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix},$$

row-reducing

$$\left[\begin{array}{cc|cc} 5 & 2 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right]$$

gives

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{7} & \frac{2}{7} \\ 0 & 1 & \frac{1}{7} & -\frac{5}{7} \end{array} \right].$$

Therefore,

$$\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{bmatrix}.$$

Tutorial Q2(b): Singular matrix

Knowledge used

- ▶ A square matrix has no inverse if its rows or columns are linearly dependent.
- ▶ Equivalently, a matrix is singular if its determinant is zero.

Solution

For $\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix}$, the augmented matrix is

$$\left[\begin{array}{cc|cc} 5 & 2 & 1 & 0 \\ 10 & 4 & 0 & 1 \end{array} \right].$$

Apply $R_2 \leftarrow R_2 - 2R_1$. Then row 2 on the coefficient side becomes zero:

$$\left[\begin{array}{cc|cc} 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array} \right].$$

So the rows are linearly dependent. Therefore, $\mathbf{A}$ is singular and has no inverse.

Tutorial Q2(c): Inverse of a 3×3 matrix

Knowledge used

- Use the same Gaussian elimination method for 3×3 matrices.
- Row-reduce $[\mathbf{A} \mid \mathbf{I}]$ to $[\mathbf{I} \mid \mathbf{A}^{-1}]$.

Solution

For

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix},$$

Gaussian elimination gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -9 & 2 & 2 \\ 0 & 1 & 0 & -\frac{41}{2} & 4 & \frac{9}{2} \\ 0 & 0 & 1 & -5 & 1 & 1 \end{array} \right].$$

Thus

$$\mathbf{A}^{-1} = \begin{bmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{bmatrix}.$$

Tutorial 11: Q3

Question 3

Find the following determinants:

$$(1) \begin{vmatrix} 5 & 2 \\ 1 & -1 \end{vmatrix}, \quad (2) \begin{vmatrix} 5 & 2 \\ 10 & 4 \end{vmatrix},$$

$$(3) \begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix}, \quad (4) \begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{vmatrix}.$$

Tutorial Q3: Determinants I

Knowledge used

- For a 2×2 matrix,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

- If a determinant is zero, the matrix is singular.

Solution

$$\begin{vmatrix} 5 & 2 \\ 1 & -1 \end{vmatrix} = 5(-1) - 2(1) = -7.$$

$$\begin{vmatrix} 5 & 2 \\ 10 & 4 \end{vmatrix} = 5(4) - 10(2) = 0.$$

Tutorial Q3: Determinants II

Knowledge used

- ▶ For 3×3 matrices, use cofactor expansion.
- ▶ Expanding along a row with zeros can reduce calculation.

Solution

$$\begin{vmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{vmatrix}$$

Expand along row 1:

$$= 1 [(-2)(-10) - 2(1)] + (-2) [4(2) - 1(-2)] = 18 - 20 = -2.$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{vmatrix}$$

Expand along row 1:

$$= 2[3(-3) - 2(-1)] - 1[4(-3) - 2(2)] + 1[4(-1) - 2(3)] = -14 + 16 - 10 = -8.$$

Tutorial 11: Q4

Question 4

A factory produces a slalom water ski that requires 3 labour-hours in the assembly department and 1 labour-hour in the finishing department. Assembly personnel receive \$19 per hour and finishing personnel receive \$16 per hour.

1. Write a matrix or vector product that would give the total labour cost per ski, then evaluate.
2. The factory also produces a trick water ski that requires 5 labour-hours in assembly and 1.5 labour-hours for finishing. Write a product of vectors that will give the total labour cost for this ski.
3. Labour-hour information can be combined into a single matrix $\mathbf{L} = \begin{bmatrix} 5 & 1.5 \\ 3 & 1 \end{bmatrix}$. Now suppose the company has two manufacturing plants, one in Sydney and one in Brisbane and that their hourly rates for each department are given by $\mathbf{H} = \begin{bmatrix} \$22 & \$19 \\ \$18 & \$16 \end{bmatrix}$ where the assembly and finishing departments are the rows and Sydney and Brisbane are the columns.
Identify a matrix product that would yield the total labour cost for each type of ski and evaluate.

Tutorial Q4: Labour costs

Knowledge used

- ▶ Use matrix multiplication to combine quantities and unit costs.
- ▶ A row vector times a column vector gives a scalar product.
- ▶ The order of multiplication matters for interpretation.

Solution

Total labour cost per slalom ski:

$$\begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 19 \\ 16 \end{bmatrix} = 3(19) + 1(16) = 73.$$

Total labour cost per trick ski:

$$\begin{bmatrix} 5 & 1.5 \end{bmatrix} \begin{bmatrix} 19 \\ 16 \end{bmatrix} = 5(19) + 1.5(16) = 119.$$

Tutorial Q4: Matrix product interpretation

Knowledge used

- ▶ Matrix multiplication can be conformable but not economically meaningful.
- ▶ Always check whether the resulting entries have the intended interpretation.

Solution

$$LH = \begin{bmatrix} 5 & 1.5 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 22 & 19 \\ 18 & 16 \end{bmatrix} = \begin{bmatrix} 137 & 119 \\ 84 & 73 \end{bmatrix}.$$

This product gives total labour costs by ski type and plant.

However,

$$HL = \begin{bmatrix} 22 & 19 \\ 18 & 16 \end{bmatrix} \begin{bmatrix} 5 & 1.5 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 167 & 52 \\ 138 & 43 \end{bmatrix}.$$

Although this product is conformable, it does not relate costs to the correct skis or plants in a meaningful way.

Key takeaways

- ▶ A linear system can be written as $\mathbf{Ax} = \mathbf{b}$.
- ▶ Gaussian elimination uses elementary row operations to solve systems.
- ▶ If $\mathbf{A}^{-1}$ exists, then $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.
- ▶ A square matrix is non-singular if and only if its determinant is nonzero.
- ▶ Cofactor expansion and determinant properties help identify whether inverses exist.